

the other hand, every individual pixel in an OLED panel generates accurate colours by exciting the phosphors to emit light of desired colour. Since this approach is similar to that of CRT TVs, OLEDs are better at producing excellent contrast levels and rendering accurate colours with rich saturation.

### Better Blacks

Being emissive displays, OLEDs don't require backlighting, which is the primary cause of black crush, or the inability to render black detail in LED panels. OLEDs can achieve perfect black levels because these panels can switch individual pixels off completely. The lack of a backlight fouling with pixels that are supposed to be darkened, affords much better black detail. Full-matrix backlit LED TVs advertised as locally-dimmed panels try to get around this problem, but they end up introducing the distracting halo effect in the darker scenes.

### No Backlight Bleed

Backlight bleed, as the name suggests, is a typical problem faced by transmissive displays such as LCDs and LED-backlit LCDs. Conspicuous bright and dark spots spread randomly across the panel are tell-tale signs of this phenomenon affecting virtually all LED televisions. Such bleeding affects colour and picture uniformity, and is caused by a range of factors ranging from imperfect backlight diffusion panels to chassis deformation. Since backlight bleed is linked to material and build quality, it is less prevalent in high-end LED panels. But because it affects cheaper displays the worst, it makes their already poor grey-scale performance end up looking even worse off. OLED TVs, on the other hand, don't suffer from these gremlins. These emissive panels don't have to bother with backlights and complicated diffusion panels like LCD panels, thereby making them completely immune to these issues.

### Faster Response Time

LCD displays have a much slower pixel response time as compared to OLEDs, which are ten times as fast as conventional LED displays. Pixel response time represents the delay in transitioning from grey to black and white to black states by individual pixels. LEDs are marked by slow response times because the liquid crystals are reconfigured by electrical stimulation to modulate the backlight. The pixels are slow to refresh because the liquid crystals can't change states fast enough, thereby causing fast moving images to appear blurred to the human eye. Slow response time can be an issue if you watch a lot of sports, action movies, or play video games. Spend more on LED televisions and you'll find fancier models that employ electronic wizardry such as backlight scanning, higher refresh rates, frame interpolation, and pixel overdrive to mitigate the issue. Interestingly, these can also be applied to OLEDs for even better blur reduction.

### Better Viewing Angles

Compared to OLED panels, LEDs are significantly more complicated. This complex design also results in the optimal viewing angle being restricted to a narrow sweet spot directly in front of the display. That's

because transmissive LCD panels pass light through various layers of colour and polarising filters in addition to a liquid crystal layer that block, bend, and vary the intensity of light to produce images. It's the resulting gap between the backlight and the various layers that introduces colour shift and other picture quality imperfections when viewed from extreme angles. Newer LCD technologies such as IPS panels have greatly improved off-axis screen performance, but nothing comes close to the viewing angles achieved by OLEDs. Because these panels emit light – as opposed to LEDs blocking it – they offer excellent off-axis performance, with virtually no colour and gamma uniformity issues even when viewed from extremely acute angles.

It's amply clear that OLED displays are worthy successors to the legacy of plasma and CRT technology. It's overcome the problems of size, weight, and power inefficiency of its emissive predecessors and in some aspects even improves upon the picture quality. Unfortunately, the efforts of the LCD lobby, which obviously wants to safeguard its multi billion dollar investment in manufacturing plants, could see the technology meet the same fate as plasma televisions. The fact that all major TV manufacturers are pushing the more expensive, yet inferior Quantum Dot LCD technology as a replacement to OLED TVs is quite concerning. Mobile phones is the only space where OLEDs are being pushed thanks to the efforts of manufacturers such as Samsung, which regularly outfits its flagship offerings such as the Galaxy S7 with AMOLED displays. The fact that Apple itself has recently ordered 100,000 OLED panels from Samsung for its iPhone is another good sign.

### A Matter of Resolution

Resolution has different ramifications depending upon the type of application. IPS Monitors generally ship in wider screen size variation with generous resolutions such as 2560 x 1440 instead of 1920 x 1080 for 27-inch models and 2560 x 1600 for 30-inch models. But before you make that sort of commitment, make sure that your gaming rig is good enough to handle the extra load. As a thumb rule, Full HD resolution is good enough for TV usage. You don't want to get 4K TVs yet because there isn't any real 4K content and existing Full HD streaming and Blu-ray content will look worse on a non-native 4K screen.

Apart from the usual trappings of OLED and LCD, panels also come in glossy and matte choices. The glossy variants are come in handy if you are keen on maintaining blacks fidelity and prefer your colours popping out unhindered. On the down side, these panels tend to be extremely susceptible to glare and reflections. Modern glossy plasma panels compensate by employing high-tech anti-reflective and anti-glare coatings, though. Matte displays, on the other hand, may reduce glare and reflections, but an aggressive matte coating can affect sharpness and diminish the integrity of whites, as well as hinder black levels. It's wiser to opt for glossy displays if you have ideal lighting conditions and no instances of glare and reflections. **d**